Kiernan X. Jennings

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EDUCATION

University of Wisconsin-Madison 2024 - Present
3rd Year Ph.D. Student in Chemical Engineering, Minor in Industrial and Systems Engineering
Advisors: Victor M. Zavala and Styliani Avraamidou
Research Area: Modeling and Large-Scale Optimization

University of Connecticut 2020 - 2024
B.S. in Chemical Engineering (Cum Laude), Minor in Computer Science

EXPERIENCE

University of Wisconsin-Madison 2024 - Present
Graduate Research Assistant
- Developed a multi-scale MILP optimization framework to analyze electrolysis devices participating in power markets across 20-year horizon at hourly resolution
- Developed capacity expansion model for grid-flexible electrolyzers, leveraging graph-based embedded surrogate models in Plasmo.jl to reduce subproblem complexity

PSOR Lab, University of Connecticut 2023 - 2024
Undergraduate Researcher
- Developed acausal modeling of wastewater treatment and anaerobic digestion using ModelingToolkit.jl and implemented models for EAGO, an open-source global optimization program written in Julia
- Developed advanced discrete event handling algorithms for PID controller in ModelingToolkit

RESEARCH OUTPUT

Publications

- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Decomposing a Multi-Scale Optimization Framework for Grid-Integrated Electrolysis using Aggregate-Informed Benders. *arXiv preprint arXiv:2607.21440* (2026)
- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. A Multi-Scale Optimization Framework for Grid-Integrated Electrolysis. *arXiv preprint arXiv:2607.09512* (2026)

Oral Presentations

- P. Brahmbhatt, **K.X. Jennings**, V.M. Zavala and S. Avraamidou. Decision-Focused Surrogate Modeling for Mixed-Integer Nonlinear Programs Via Neural Differentiable Optimization Layers. *2026 AIChE Annual Meeting*, Boston, MA.
- P. Brahmbhatt, **K.X. Jennings**, A. McCullough, V.M. Zavala, and S. Avraamidou. Multi-Market Bidding and Demand Response for Redox Reservoir-Based Electrochemical Devices Using Two-Stage Stochastic Optimization. *2025 AIChE Annual Meeting*, Boston, MA. [link]

Poster Presentations

- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Multi-Scale Optimization for Capacity Planning and Operations of Electrolysis Systems. *TWCCC 2025*, Madison, WI. [link]

- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Multi-Scale Optimization for Evaluating Electrolyzer Operation Degradation in Electricity Markets. *Power Grid-Informed Electrochemistry Conference 2025*, Madison, WI. [link]

MENTORING & TEACHING EXPERIENCE

Advised Students

- Emilia Veum — May 2026-Present
  (Project: Carbon Aware Grid-Integrated Electrolysis Optimization)
- Aayana Roy — Jun 2025-May 2026
  (Project: Capacity Expansion Modeling of UW-Madison Campus)

Teaching Experience

- Teaching Assistant, Process Design Senior Capstone, UW-Madison — Fall 2025, Spring 2026

SKILLS & COURSES TAKEN

Skills

- Optimization (JuMP.jl, Plasmo.jl)
- Control Systems (Simulink)
- Process Simulation (AspenPlus)
- Programming (Julia, Matlab, Python, C)

Courses

- CBE 660 (Advanced Mathematics)
- CBE 710 (Advanced Thermodynamics)
- CBE 735 (Kinetics and Catalysis)
- CBE 750 (Advanced Chemical Process Synthesis and Optimization)
- CBE 781 (Biological Engineering)
- ISyE 524 (Introduction to Optimization)
- ISyE 726 (Nonlinear Optimization I)
- ISyE 823 (Mixed-Integer Nonlinear Optimization)

AWARDS & SERVICE

Outreach Services

- University Engineering Expo Volunteer — 2024-Present
- Circular Ecnomies Systems Group Climate & Inclusion Officer — 2024-Present
- Swim Across the Sound Group Leader — 2022-2024
- HuskyTHON Student Group Leader — 2021-2024

Academic Awards

- Sunny and Po Lo Graduate Fellowship Award — 2024-2025
- Homer Babbidge Scholar — 2021-2022
- University's Dean List — 2021-2024
- Academic Excellence Award — 2020-2024